CloudOps Challenge

Hackathon

TalenciaGlobal 11/28/24 Cloud and DevOps

Contents

GDB CloudOps Challenge 2026

From Code to Cloud: Launching a Digital Bank

Story

Global Digital Bank (GDB) is a rapidly growing digital-first banking institution preparing to launch its next-generation banking platform.
Over the past year, the engineering team has successfully developed a modern microservices-based banking ecosystem comprising customer onboarding, authentication, account management, transaction processing, payment services, and enterprise administration.
The application has passed functional testing and is ready for deployment. However, there is one critical challenge.
The entire platform currently runs only on developer machines.
The Board of Directors has announced a launch date, and the CloudOps Team has been tasked with transforming the platform into a production-ready cloud-native solution.
As members of the GDB CloudOps Engineering Team, your mission is to build the deployment ecosystem, automate delivery pipelines, deploy services, configure cloud infrastructure, and ensure the bank is ready for launch.
The countdown has begun.
Can your team successfully launch Global Digital Bank before the deadline?

Background

The GDB platform consists of:

Frontend Application

- Global Digital Bank React Application

Infrastructure Services

- Eureka Server
- API Gateway

Business Services

- Auth Service
- Users Service
- Account Service
- Transactions Service
- Payment Gateway Service
- Company Service
- Aadhaar Service

Databases

- gdb_auth_db
- gdb_users_db

- gdb_accounts_db
- gdb_transactions_db

Technology Stack

- Java Spring Boot
- React 18 + Vite
- PostgreSQL
- Docker
- Git
- GitHub/GitLab
- AWS/Azure/GCP
- Kubernetes (Bonus)
- Flyway Database Migration

Goal

Your objective is to deploy the entire Global Digital Bank ecosystem from source code to a fully operational cloud environment using industry-standard DevOps practices.
Participants must:

- Establish Git workflows
- Containerize applications
- Deploy locally using Docker
- Provision cloud infrastructure
- Create CI/CD pipelines
- Deploy services to cloud environments
- Configure service communication
- Validate end-to-end functionality

Level 1: Git Foundation

Title

Version Control Governance

Description

Establish a professional Git workflow that supports collaborative development, release management, and production deployments.

Tasks

1. Create repository structure
2. Create main branch
3. Create dev branch
4. Create feature branches
5. Implement pull request workflow
6. Configure branch protection rules
7. Document branching strategy

Deliverables

- Git Repository

- Branching Strategy Document
- Pull Request Evidence

Success Criteria

- main branch created
- dev branch created
- Feature branch workflow implemented
- Pull request completed
- Branch protection enabled

Level 2: Containerization Challenge

Title

Containerize the Bank

Description

Package all applications and services as Docker containers to ensure consistency across environments.

Tasks

1. Create Dockerfile for React Application
2. Create Dockerfile for Eureka Server
3. Create Dockerfile for API Gateway
4. Create Dockerfile for all Microservices
5. Build Docker images
6. Validate startup

Deliverables

- Dockerfiles
- Docker Images
- Build Instructions

Success Criteria

- All services build successfully
- Containers start successfully
- No build failures
- Images tagged correctly

Level 3: Local Integrated Deployment

Title

Bank in a Box

Description

Deploy the complete banking ecosystem locally using Docker Compose.

Tasks

1. Create PostgreSQL containers
2. Configure databases
3. Deploy Eureka Server
4. Deploy API Gateway
5. Deploy all microservices
6. Deploy frontend application
7. Configure networking

Deliverables

- docker-compose.yml
- Environment Variables
- Deployment Guide

Success Criteria

- All containers running
- Eureka operational
- Services registered
- Gateway operational
- Frontend accessible

Level 4: Cloud Infrastructure Setup

Title

Build the Banking Cloud

Description

Provision cloud infrastructure capable of hosting the banking platform.

Tasks

1. Create cloud project/account
2. Provision VM instances
3. Configure networking
4. Configure firewalls/security groups
5. Configure SSH access
6. Configure storage
7. Configure DNS

Deliverables

- Infrastructure Diagram
- Cloud Screenshots

- Access Documentation

Success Criteria

- Cloud environment provisioned
- Connectivity established
- Security rules configured
- Infrastructure accessible

Level 5: CI/CD Pipeline Engineering

Title

Automate the Release Process

Description

Create automated pipelines that build, test, package, and publish applications.

Tasks

1. Integrate Git Repository
2. Trigger builds on commits
3. Build Java services
4. Build React application
5. Build Docker images
6. Push images to registry

Deliverables

- Pipeline Definition
- Build Logs
- Registry Screenshots

Success Criteria

- Automated builds
- Automated Docker image creation
- Successful registry push
- Pipeline execution evidence

Level 6: Cloud Deployment

Title

Go Live Deployment

Description

Deploy the entire banking ecosystem to the cloud environment.

Tasks

1. Deploy PostgreSQL databases
2. Deploy Eureka Server
3. Deploy API Gateway
4. Deploy all microservices
5. Deploy frontend
6. Configure startup sequence

Deliverables

- Deployment Scripts
- Deployment Documentation

Success Criteria

- All services deployed
- Services reachable
- Databases connected
- Frontend deployed

Level 7: Configuration & Integration

Title

Connect the Ecosystem

Description

Configure all application endpoints and validate communication across services.

Tasks

1. Configure Eureka URL
2. Configure Gateway Routes
3. Configure Database Connections
4. Configure Frontend API URLs
5. Configure Environment Variables
6. Configure Service Discovery

Deliverables

- Configuration Files
- Architecture Diagram

Success Criteria

- Services communicate successfully
- Gateway routing operational
- Database connectivity verified
- Frontend integrated

Level 8: Production Validation

Title

Launch Readiness Verification

Description

Validate that the entire banking platform is operational and production-ready.

Tasks

Authentication

- Login
- JWT Token Generation

User Management

- Create User
- Update User

Account Management

- Create Account
- View Account

Transactions

- Deposit
- Withdrawal
- Transfer

Infrastructure

- Eureka Validation
- Gateway Validation

Deliverables

- Test Report
- Evidence Screenshots
- API Test Collection

Success Criteria

- Frontend operational
- APIs operational
- Service discovery working
- Gateway routing working
- Database updates successful

Bonus Level 9: Kubernetes Modernization

Title

Cloud-Native Banking Platform

Description

Transform the deployment into a scalable Kubernetes-based platform.

Tasks

1. Create Namespace
2. Create Deployments
3. Create Services
4. Create ConfigMaps
5. Create Secrets
6. Create Ingress
7. Configure Autoscaling
8. Rolling Updates

Deliverables

- Kubernetes Manifests
- Deployment Evidence

Success Criteria

- Pods healthy
- Services exposed
- Ingress operational
- Autoscaling configured

Final Mission Success

A team is considered to have successfully launched Global Digital Bank when:

- Source code is managed through Git workflows
- Applications are containerized
- CI/CD pipelines are operational
- Infrastructure is deployed in the cloud
- Services communicate through Eureka and API Gateway
- Databases are connected
- Frontend is accessible
- Banking operations execute successfully